Motor Vehicle Theft in New Zealand

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Research Question: What age group and geographic region are most commonly associated with motor vehicle theft and related offences in New Zealand, and how have the offender counts for these crimes evolved between 2019 and 2023 within these age group and geographic region?

Introduction

The recent surge in motor vehicle theft and related offences in New Zealand has become a significant concern, with media reports highlighting a notable increase in such crimes (RNZ, 2024; The Post, 2023). These related offences include illegal use of motor vehicles, theft of vehicle parts or contents and the theft of items from within vehicles. This has led to increased costs for repairs, replacements and insurance premiums, as well as causing emotional distress for the car owner. The purpose of this research is to explore the trends and patterns associated with motor vehicle theft and related offences in New Zealand. By identifying the specific age group and geographic region most frequently involved, as well as examining how offender counts for these crimes have evolved between 2019 and 2023 within these age group and geographic region, this study aims to provide law enforcement agencies with valuable data or insights to develop more effective targeted prevention strategies. Ultimately, the goal is to enhance public safety and improve the quality of life for residents by reducing the occurrence of these crimes.

Methods

A total of 60 monthly unique offenders (demographics) datasets were downloaded from the NZ Police website (New Zealand Police, 2024). To access these datasets, click on "District" and choose the desired month and year from the "12-month period ending" dropdown menu. Then, locate the Download icon at the bottom right corner of the section and select the "Data" option. This action will open a new Tableau window. In the tab section, click on "Full Data", then select "Show Fields" and check "(All)" option. Finally, proceed to download. Since only one monthly dataset can be downloaded at a time, these 60 datasets, covering the period from January 2019 to December 2023, were downloaded manually, one by one. Once completed, the datasets were merged into a single dataset containing 2,759,421 observations and 22 variables. To focus on the specific crimes of interest, the dataset was filtered to include only observations with the ANZSOC subdivision "Motor Vehicle Theft and Related Offences," reducing it to 68,934 observations. From this filtered data, 5 key variables were selected and retained for analysis: ANZSOC subdivision, date (month and year), age group, police district, and offender count.

Since this research focuses on the 5-year period from 2019 to 2023, the date variable was preprocessed to aggregate monthly data into yearly intervals. Then, the offender counts will be grouped by year, age group or police district to create new data frames, each containing the total number of offenders for that specific grouping. These data frames will be used for further analysis and visualization. To explore the data, two bar charts were generated: one to display the total offender counts across different age groups, and the other to show offender counts by police district. Bar charts were chosen for its effectiveness in illustrating categorical data, allowing for easy comparison of the offender counts across different age groups and police districts. To capture trends over time, three separate line charts were created, visualizing the total offender counts, total offender counts by age group and total offender counts by police district over the five-year period from 2019

to 2023. Line charts were chosen for their ability to effectively display time series data, making it easier to identify trends and patterns over multiple years.

The types and roles of the 5 key variables are described as follows:
1. ANZSOC subdivision
Type: Independent Nominal Variable
Role: It is a part of the national statistical framework used to classify criminal behaviour in Australia and New Zealand. It includes various crime categories. For this analysis, motor theft and related offenses have been filtered and selected, allowing us to focus on this specific crime type and investigate how it varies across different years, age groups and police districts.

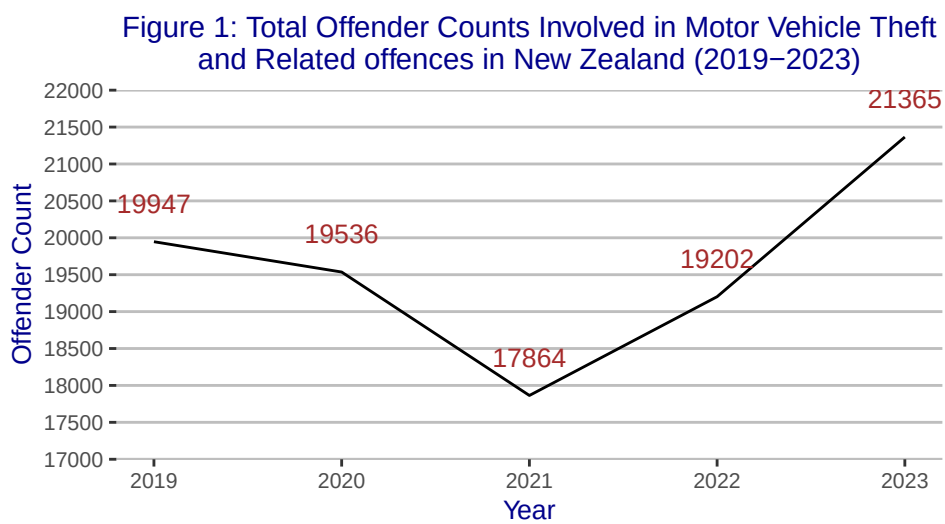
2. Year
Type: Independent Nominal Variable
Role: It is used to track changes in offender counts over time, helping to understand whether the offender counts fluctuate across different years.

3. Age Group
Type: Independent Nominal Variable
Role: It categorizes the offenders based on their age, allowing us to investigate whether certain age groups are more or less likely to be involved in motor vehicle theft and related offences.

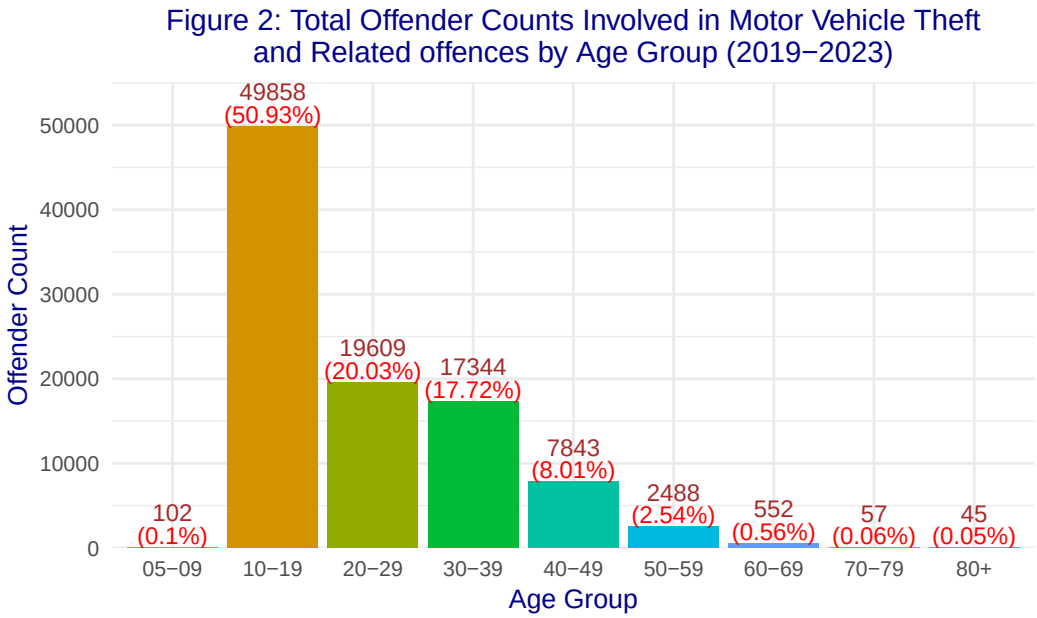
4. Police District
Type: Independent Nominal Variable
Role: It represents the geographic location where the offences occurred. By examining offender counts across different police districts, the areas with higher or lower offender rates of motor vehicle theft and related offences can be identified.

5. Offender Count
Type: Dependent Discrete Variable
Role: It serves as the outcome variable in this analysis, representing the number of individuals involved in motor vehicle theft and related offences. Offender counts will be aggregated to compute the total offender count, based on year, age group or police district, depending on the specific analysis required.

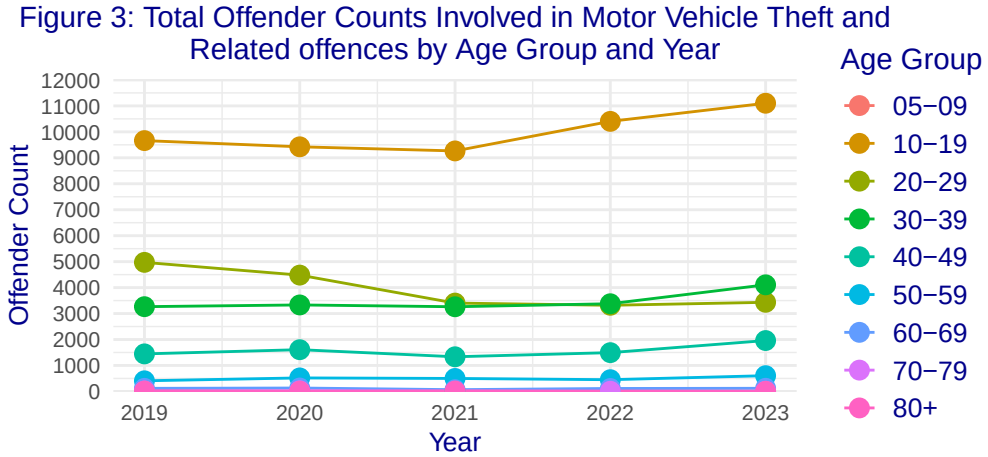
Results



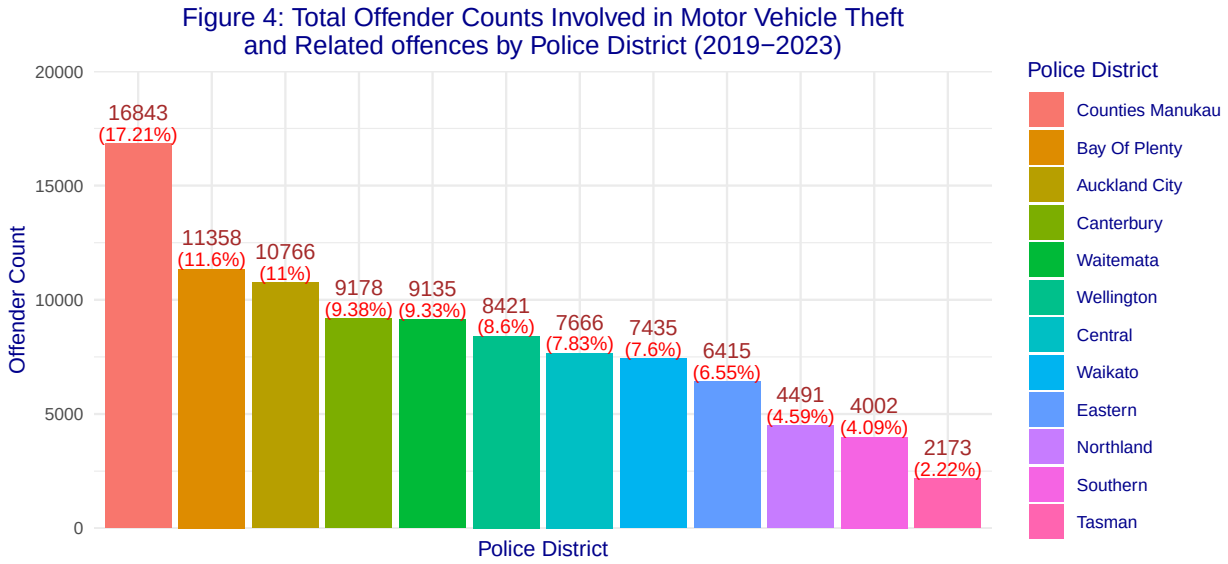
In figure 1, the values represent the total offender counts grouped by year. The line chart shows a fluctuating trend in motor vehicle theft and related offences between 2019 and 2023. The total number of offenders decreased from 19,947 in 2019 to 17,864 in 2021. However, this downward trend reversed in 2022, with a sharp increase to 19,202. This upward trajectory continued in 2023, reaching a total of 21,365 offenders.



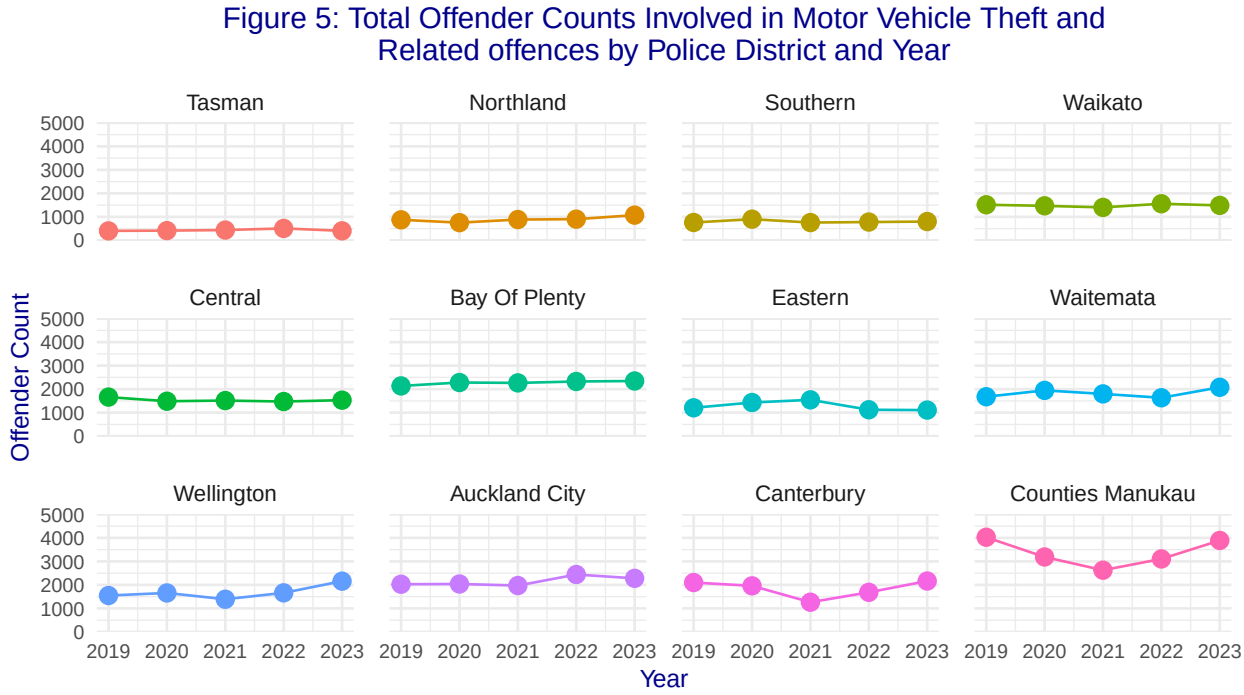
In figure 2, the values represent the total offender counts grouped by age group. The bar chart reveals that the 10-19 age group accounts for 50.93% of all offenders, making the most represented group involved in motor vehicle theft and related offences between 2019 and 2023. This trend generally decreases as age increases, with the exception of the 05-09 age group, which represents a negligible 0.1% of the total offender population.



In figure 3, each point represents the total offender counts grouped by age group and year. The exact values for each point can be found in Table 1 in the appendix section. The line chart shows that the 05-09, 50-59, 60-69, 70-79, and 80+ age groups exhibit a relatively stable trend, with minimal variation in offender counts over the five-year period. However, the figure 3 cannot clearly distinguish between the trends for age-group 5-09, 60-69, 70-79, and 80+ due to their similiar offender counts and graph limitations (the exact values can be found in Table 1). The 20-29 age group shows a noticeable decline from 2019 to 2022, then remains steady from 2022 to 2023. In contrast, the 10-19, 30-39, and 40-49 age groups experience a decreasing trend from 2019 to 2021, followed by a marked upward trend from 2021 to 2023, with the 10-19 group experiencing a significantly sharper increase compared to the other two groups.



In figure 4, the values represent the total offender counts grouped by police district. The bar chart shows that the Counties Manukau police district has the highest offender count, accounting for 17.21% of all offenders across districts. This is followed by the Bay of Plenty with 11.6% and Auckland City at 11%. In contrast, the Tasman Police District has the lowest number of offenders at 2.22%. It is closely followed by the Southern (4.09%) and Northland (4.59%) districts.



In figure 5, each point represents the total offender counts grouped by police district and year. The exact values for each point can be found in Table 2 in the appendix section. The line chart shows that 6 police districts – Tasman, Northland, Southern, Waikato, Central, Bay of Plenty – exhibited a relatively stable trend, with minimal variation in offender counts over the five-year period. In the Eastern police district, there was a significant upward trend from 2019 to 2021, followed by a decline from 2021 to 2023. In the Waitemata police district, offender counts increased from 2019 to 2020, then declined from 2020 to 2022,

before experiencing a noticeable rise in 2023. The Wellington police district experienced a slight increase from 2019 to 2020, followed by a decrease in 2021, before experiencing a marked increase from 2022 to 2023. In Auckland City, the trend remained stable from 2019 to 2021, then rose noticeably from 2021 to 2022 before declining from 2022 to 2023. Both the Canterbury and Counties Manukau police districts showed a significant decrease in offender counts from 2019 to 2021, but experienced a noticeable increase from 2021 to 2023, with Counties Manukau in particular seeing a substantial rise in total offender counts during this period.

Discussion

Age Group and Police District Analysis

The results from figures 2 and 4 indicate that the age group 10-19 and the Counties Manukau police district are most commonly associated with motor vehicle theft and related offences in New Zealand. Young people in the 10-19 age group often engage in these crimes as thrill-seeking activities for excitement, or to gain social status among their peers. Their behavior is sometimes amplified by the desire to showcase their activities on social media platforms. Moreover, many young people in the 10-19 age group are either still in school or have limited employment opportunities. Economic hardships, especially when coupled with a lack of family support, may drive them to engage in criminal activities as a source of earning money. Therefore, these factors likely contribute this age group having highest offender count for motor vehicle theft and related offences compared to other age groups. Counties Manukau, being one of the most populous and rapidly growing districts in New Zealand, also has the highest offender count for motor vehicle theft and related offences compared to other districts. The correlation between high population density and increased motor vehicle theft and related offences is evident. The district's urbanized nature, with its large number of vehicles parked on streets or in easily accessible parking locations, provides more opportunities for motor vehicle theft and related offences to occur.

Trend Analysis

The 10-19 age group and the Counties Manukau police district shows a decline in total offender count for motor vehicle theft and related offences from 2019 to 2021, followed by an increase from 2021 to 2023 as shown in figures 3 and 5. This pattern aligns with the overall trend in motor vehicle theft and related offences shown in figure 1. Similar trends are also observed in specific age groups (30-39, 40-49) and police districts (Auckland and Canterbury). The reduction in offences between 2019 and 2021 can largely be attributed to the Covid-19 pandemic. During this period, lockdowns and travel restrictions were enforced, leading to fewer cars being parked in public spaces as people stayed home. With vehicles kept in secured locations, such as garages or driveways, opportunities for motor vehicle theft and related offences were significantly reduced. However, the rise in offences from 2021 to 2023 can be explained by the easing of restrictions and the return to normal activities. This led to greater mobility and more vehicles being parked in public spaces, providing more opportunities for theft. Additionally, post-pandemic economic challenges, including rising unemployment and inflation, have further strained communities financially, contributing to the resurgence of motor vehicle theft and related offences.

Recommendations

The 10-19 age group and the Counties Manukau police district have been identified as having a high prevalence of motor vehicle theft and related offences. To address this issue, targeted prevention programs should be implemented. For the 10-19 age group, the government should invest in and develop after-school programs, sports and recreational activities to divert them from engaging in criminal behavior. Additionally, job training and vocational programs should be offered for young people struggling in traditional educational settings. By equipping them with job-related skills and increasing their employment opportunities, it will ease their financial stress and decrease the likelihood of engaging in motor vehicle theft and related offences. To address the high offender count of motor vehicle theft and related offences in Counties Manukau district, the government should allocate police resources more effectively by increasing police patrols in high-risk areas and utilizing data analytics to identify specific hotspot within the district. By focusing efforts on these areas, law enforcement can significantly improve the chances of apprehending offenders. Additionally, collaborating with the local council to raise public awareness about the importance of anti-theft devices, such as GPS trackers, immobilizers and alarm systems, and potentially subsidizing these technologies for residents in high-theft area could further reduce the motor vehicle theft and related offences. Finally, given the general trend of

increased motor vehicle theft and related offences following the Covid-19 pandemic, likely driven by inflation and unemployment, the government should also strengthen welfare programs, such as unemployment benefits, housing support and food assistance. This will ease the economic pressures that may drive individuals toward criminal activity, thereby reducing the likelihood of motor vehicle theft and related offences.

References

New Zealand Police. (2024). *Unique Offenders (demographics)* [Dataset]. Retrieved September 10, 2024, from https://www.police.govt.nz/about-us/publications-statistics/data-and-statistics/policedatanz/unique-offenders-demographics

RNZ. (2024, June 27). *Car thefts, online fraud sharply on the rise – Ministry of Justice survey.* https://www.rnz.co.nz/news/national/520681/car-thefts-online-fraud-sharply-on-the-rise-ministry-of-justice-survey

The Post. (2023, December 9). *Vehicle stolen? Good luck getting it back.* https://www.thepost.co.nz/nz-news/350110984/vehicle-stolen-good-luck-getting-it-back

Appendix

Table 1: Total Offenders Involved in Motor Vehicle Theft and Related offences by Age Group and Year

	2019	2020	2021	2022	2023
05-09	42	5	14	31	10
10-19	9663	9424	9265	10404	11102
20-29	4972	4481	3404	3319	3433
30-39	3265	3332	3262	3378	4107
40-49	1446	1608	1335	1494	1960
50-59	409	520	500	452	607
60-69	119	131	67	114	121
70-79	16	27	0	7	7
80+	15	8	17	0	5

Table 2: Total Offenders Involved in Motor Vehicle Theft and Related offences by Police District and Year

	2019	2020	2021	2022	2023
Tasman	404	415	439	509	406
Northland	875	754	886	902	1074
Southern	761	902	758	780	801
Waikato	1515	1470	1403	1559	1488
Central	1662	1486	1514	1473	1531
Bay Of Plenty	2137	2282	2266	2326	2347
Eastern	1206	1432	1545	1123	1109
Waitemata	1678	1947	1798	1633	2079
Wellington	1549	1658	1391	1663	2160
Auckland City	2029	2040	1973	2443	2281
Canterbury	2102	1961	1264	1685	2166
Counties Manukau	4029	3189	2627	3106	3892